

Starting Finance Club PoliTo

Research Paper

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RESEARCH ARTICLE

Operating-Value Sensitivity for Barrick Mining

A controlled comparison of option-implied gold-price laws

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JEL classification

G12; G13; G31; C32; C53; C61; C63

Discoveries

Calibrated engines share one unreconciled operating contract.

Black–Scholes		60.10
Heston		59.77
Bates–Poisson		59.12
Bates–Hawkes		59.82

Signed-terminal proxy; USD billion.

Sensitivity only; not company value or a physical probability forecast.

Abstract

This study audits the sensitivity of an unreconciled Barrick operating-value proxy to alternative option-implied gold-price laws. Eight student teams connect probability, econometrics, simulation, operating forecasts and a common discounted-margin mapping. Black–Scholes, Heston, Bates and Bates–Hawkes are fitted to the same 64 GLD calls dated 2 September 2026. A calibration audit retains feasible nested candidates and improves the previously reported Bates and Hawkes fits; historical out-of-sample scores remain a separate, frozen experiment. Production, unit costs and annual discount-rate shocks are shared across 8,192 simulated paths. The reported quantities are aggregate operating proxies under explicit terminal conventions: no net-debt, ownership or diluted-share bridge is asserted. Option-implied risk-neutral dynamics combined with an assumed WACC do not constitute a validated corporate valuation or physical forecast. Signed perpetuity and five-year-only closures expose the sensitivity to terminal assumptions. Numerical checks also show that small cross-model differences in medians are not stable evidence of economic ranking. The contribution is a reproducible, controlled research experiment with explicit limits on accounting, probability measure, option conventions, extrapolation and statistical inference. Results are neither fair values nor price targets, investment recommendations or evidence of market mispricing.

RESEARCH PROGRAMME AND COLLABORATION

This paper grew out of an eight-team project involving 17 students with complementary backgrounds in mathematics, data science, engineering, economics and finance. The size of the author group reflects a division of research responsibilities and a shared learning process: Teams 1–3 developed the mathematical, econometric and simulation foundations; Teams 4–5 built the operating and corporate-valuation layers; and Teams 6–8 studied gold-price dynamics, market evidence and option-implied calibration.

The extended [technical report](#) records definitions, derivations and implementation choices to establish a common technical language across these backgrounds. Teams worked on dedicated modules, then compared assumptions, revised their contributions and reconciled dates, units and interfaces before integration. The present paper condenses this work into a single auditable experiment. Together with the reproducible code, it demonstrates the group's existing research practice and provides a basis for developing a sustained student-led quantitative research team with academic guidance.

1 INTRODUCTION

A gold producer's value depends on the distribution of future commodity prices as well as its operating and financing assumptions. A single price path can hide this source of model risk. We ask how an unconstrained Barrick operating-proxy distribution changes when progressively richer gold-price processes are substituted within a common operating scenario and discounted-margin contract.

We compare Black–Scholes, Heston, Bates–Poisson and Full Bates–Hawkes calibrated to GLD options. Current-surface diagnostics and rolling out-of-sample tests distinguish fit from predictive performance; shared production, costs and WACC shocks isolate the effect of the gold engine. The contribution is this controlled comparison and its auditable integration, rather than a claim that any model provides an unconditional fair value.

2 DATA AND VALUATION SETUP

2.1 Controlled layer separation

The integrated experiment is organized as a controlled handoff across three modules. Team 8 governs only the gold-price process; Team 4 supplies the operating forecasts; and Team 5 maps those inputs through a common tax, WACC, terminal-value and equity-bridge contract. In the four-model valuation comparison, only the gold engine changes. Barrick's Q1–Q2 2026 actuals and the same Team 4 production and unit-cost forecasts from Q3 2026 onward are frozen over 20 quarters; the same Team 5 corporate rules and WACC shocks are then reused across runs.

2.2 Calibration snapshot and measure discipline

The current calibration uses the 2 September 2026 GLD call surface acquired through the IB API. After the common DTE ≥ 75 and numerical filters, the dense surface contains 605 eligible observations across 12 expiries and 146 distinct strikes, with 79–653 days to expiry. The official 8×8 Chebyshev–Chebyshev sample contains 64 distinct actual contracts spanning eight expiries and 20 strikes. The risk-free input is a same-date NSS fit to 14 U.S. Treasury tenors ([Svensson, 1994](#); [U.S. Treasury, 2026](#)). Its 2.0608 bp RMSE is computed against the continuously compounded par-yield proxy; the curve is not described as a bootstrapped zero curve. The Barrick market reference is the 2 September close of USD 44.13, while the common gold anchor remains Barrick's Q2 2026 average realized price of USD 4,417/oz ([Barrick, 2026](#)). The operating calendar is a fixed scenario index, not a forecast commencing on 2 September; earlier actuals are not future observations.

The option layer is estimated under the risk-neutral measure $\mathbb{Q}$. In the corporate experiment, only the option-implied distributional shape is transferred to the realized-price anchor. No GLD-share-to-ounce conversion or validated $\mathbb{Q} \rightarrow \mathbb{P}$ transformation is claimed. Combining this law with WACC defines a sensitivity operator, not risk-consistent pricing.

Table 1. Frozen contract used for model attribution.

Block	Current experiment
Gold layer	Black–Scholes/GBM, Heston, Bates–Poisson and Full Bates–Hawkes calibrated by Team 8 under $\mathbb{Q}$.
Operations	Barrick Q1–Q2 2026 actuals followed by Team 4 deterministic production and unit-cost vectors from Q3 2026.
Corporate mapping	Team 5 tax, growth–ROIC–reinvestment schedule, stochastic WACC, terminal-value and equity-bridge logic.
Simulation	8,192 paths, 260 fine steps, 20 quarters and a common WACC shock array.

2.3 Option exercise convention

GLD contracts are American-style (Cboe, 2026). We set payout yield $q = 0$: trust expenses reduce backing, not cash distributions (SPDR, 2025). For non-distributing calls and positive rates, early exercise is suboptimal in the frictionless model. An 800-step CRR check on all 64 contracts gives an American–European premium below 10^{-8} USD; this tests that convention, not market frictions.

2.4 Operating-model status

Team 4’s original operating study is richer than the object ultimately passed downstream. Cost of Sales is forecast from a log-space master chain; production is constructed bottom-up from ore processed, average grade and recovery. The unified valuation refreshes the handoff with 2026 actuals but still freezes the resulting 20-quarter production and cost vectors. Consequently, mine-level operating uncertainty is not propagated jointly with gold-price uncertainty. Team 4’s legacy price simulation and illustrative EBITDA/valuation output are explicitly excluded.

3 METHODOLOGY

3.1 Gold-price engines

The four Team 8 specifications relax one restriction at a time. Under $\mathbb{Q}$, Black–Scholes/GBM uses constant volatility,

$$\frac{dS_t}{S_t} = (r_t - q_t) dt + \sigma dW_t. \quad (1)$$

GBM is the constant-volatility benchmark. Heston introduces mean-reverting stochastic variance (Heston, 1993),

$$\frac{dS_t}{S_t} = (r_t - q_t) dt + \sqrt{v_t} dW_t^S, \quad (2)$$

$$dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^v, \quad dW_t^S dW_t^v = \rho dt. \quad (3)$$

Bates adds compensated compound-Poisson jumps (Bates, 1996),

$$\frac{dS_t}{S_t^-} = (r_t - q_t - \lambda k) dt + \sqrt{v_t} dW_t^S + (e^J - 1) dN_t, \quad (4)$$

where $k = \mathbb{E}^{\mathbb{Q}}[e^J - 1]$.

3.2 Full Bates–Hawkes: implemented affine engine

The full engine preserves Heston variance but replaces the constant Poisson arrival rate with an exponential Hawkes state (Hawkes, 1971):

$$\frac{dS_t}{S_t^-} = (r_t - q_t - \lambda_t k) dt + \sqrt{v_t} dW_t^S + (e^J - 1) dN_t, \quad (5)$$

$$dv_t = \kappa(\theta_v - v_t) dt + \xi \sqrt{v_t} dW_t^v, \quad (6)$$

$$d\lambda_t = \beta(\bar{\lambda} - \lambda_t) dt + \alpha dN_t, \quad n = \alpha/\beta < 1. \quad (7)$$

Under the implemented exponential-kernel specification, pricing exploits the affine Markov structure rather than a standalone finite-difference PIDE. Writing $X_t = \log S_t$, the characteristic function factorizes into Heston diffusion and Hawkes jump blocks,

$$\phi_{X_T}(u) = \phi_{\text{diff}}(u) \exp(A(u, T) + B(u, T)\lambda_0), \quad (8)$$

with the jump block determined by the Riccati system

$$\frac{dB}{d\tau} = \psi_J(u) e^{\alpha B} - \beta B - iuk - 1, \quad (9)$$

$$\frac{dA}{d\tau} = \beta \bar{\lambda} B, \quad A(0) = B(0) = 0, \quad (10)$$

where $\psi_J(u) = \exp(iu\mu_J - \frac{1}{2}\sigma_J^2 u^2)$. European call prices use Fourier-COS inversion (Fang and Oosterlee, 2008), with independent diffusion and jump drivers. The identities $\phi(0) = 1$, the martingale condition at $u = -i$, and the limit $\alpha \rightarrow 0$, $\lambda_0 = \bar{\lambda}$, back to Bates are used as implementation checks.

Model-role constraint

These four processes govern only the gold-price layer. They never generate Barrick production or costs in the integrated valuation.

3.3 Option-surface calibration and predictive test

Every model is fitted to the same structured GLD surface by minimizing a vega-scaled price loss,

$$F(\theta) = \frac{1}{N} \sum_{i=1}^N \left[\frac{C_i^{\text{model}}(\theta) - C_i^{\text{mkt}}}{\max(\text{Vega}_i, \varepsilon)} \right]^2, \quad (11)$$

with Vega per unit decimal volatility and $\varepsilon = 10^{-4}$. This is price-based MSE; exact IV RMSE uses inversion, a square root and 10^4 scaling. Nested starts are retained during SLSQP refinement. A Feller strict-positivity constraint is imposed by choice; Hawkes stability requires $n < 1$.

Calibration fit and forecasting are evaluated separately. A date is admitted to the rolling test only when at least 64 eligible observations and three expiries survive the common filters. Each CC64 origin is scored on the next dense date. With n_t common targets on date t , equal-date performance is

$$R_{OOS, m|b}^2 = 1 - \frac{\sum_t n_t^{-1} \sum_j e_{m,t,j}^2}{\sum_t n_t^{-1} \sum_j e_{b,t,j}^2}, \quad (12)$$

against both the origin cross-sectional mean IV and an origin observed-IV surface interpolated in maturity and log-moneyness inside its convex hull. The target spot and rate curve enter only as scoring-date conditioning variables. This design prevents the lowest in-sample IV error from being interpreted automatically as predictive dominance.

Origin parameters are frozen before next-date scoring; target quotes do not select them. Conditioning on target spot and rates makes this a repricing test, not a joint ex ante forecast. Persistence is evaluated only on its interpolation support; unmatched observations are not zero losses.

3.4 Team 4 cost and production models

For costs, Team 4 first builds a log-space global master chain from mine-level series and then selects an **ARIMA(3,1,0) with drift** by automated information-criterion search. The admitted specification is

$$(1 + 0.9501B + 0.6128B^2 + 0.3003B^3)(1 - B)y_t = 0.0222 + \varepsilon_t. \quad (13)$$

The selected model minimizes both $AIC = -94.64$ and $BIC = -86.87$ (Akaike, 1974; Schwarz, 1978), where p counts parameters and N observations,

$$AIC = 2p - 2 \log \hat{L}, \quad BIC = p \log N - 2 \log \hat{L}. \quad (14)$$

The source reports a 12-period hold-out MAPE of 0.603%; its sample, missing-value treatment and test details are not reproduced here, so this is not independent validation.

Team 4 models production components without claiming AIC/BIC selection. With ore in kt, grade in g/t and recovery as a fraction, physical output in koz is

$$Q_t^{\text{phys}} = \text{Ore}_t \text{Grade}_t \text{Recovery}_t / 31.1035, \quad (15)$$

with ore modelled as $SARIMA(1,0,1)(1,0,0)_4$, grade as $SARIMA(1,0,0)(0,0,0)_4$, and recovery sampled from bounded historical ranges. Team 4 additionally applies a 0.91 empirical realization

factor, not a unit conversion; its transfer across mines remains un-validated. The valuation retains the frozen Team 4 vectors, not the Goodman distributions; product-variance formulas require explicit dependence assumptions.

3.5 Valuation mapping

Team 5 makes growth internally consistent with reinvestment through

$$b_t = \frac{g_t}{ROIC_t}, \quad F_t^{proxy} = M_t^{AT} - \max(M_t^{AT}, 0)b_t, \quad (16)$$

where M_t^{AT} is the after-tax component margin. Over the five explicit years, g_t declines linearly from 8.50% to 2.50%, $ROIC_t$ from 14.50% to 10.50%, and the implied reinvestment rate from 58.62% to 23.81%.

The common discount-rate process is mean reverting,

$$w_{t+1} = \bar{w} + (w_t - \bar{w})e^{-\kappa_w} + \sigma_w^* \varepsilon_{t+1}, \quad (17)$$

with $w_0 = 9.10\%$, $\bar{w} = 8.30\%$, $\kappa_w = 0.45$ and annual volatility 0.90%. It is clipped to $[g_\infty + 1.00\%, 25.00\%] = [3.50\%, 25.00\%]$; every gold engine receives the same 8,192-path shock array.

For simulated path i , the common operating contract and discount rates define the operating proxy,

$$V_{op}^{(i)} = \sum_{t=1}^T \frac{F_t^{proxy, (i)}}{\prod_{s=1}^t (1 + WACC_s^{(i)})} + \frac{TV_T^{(i)}}{\prod_{s=1}^T (1 + WACC_s^{(i)})}. \quad (18)$$

The terminal block uses $g_\infty = 2.50\%$, $ROIC_\infty = 10.50\%$ and therefore $b_\infty = 23.81\%$:

$$TV_T^{(i)} = \frac{M_T^{AT, (i)} (1 + g_\infty) (1 - g_\infty / ROIC_\infty)}{WACC_T^{(i)} - g_\infty}, \quad (19)$$

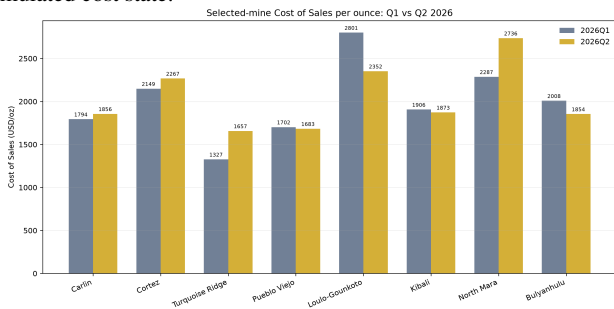
The WACC floor ensures a positive spread. A signed terminal removes the legacy zero floor; $TV = 0$ gives a five-year-only sensitivity. Neither closure is a reserve-calibrated mine life or an optimal abandonment rule. No equity/share bridge is inferred.

4 CURRENT OPERATING-MODEL EVIDENCE

Team 4 supplies the inherited operating scenarios, refreshed with Q1–Q2 2026 statistics. Its earlier gold simulation is excluded: all price paths come from Team 8. These fixed inputs permit model attribution but are not independently validated operating forecasts.

4.1 Cost-of-sales block

The cost model in Section 3.4 extracts a common log-space trend, forecasts it with ARIMA(3,1,0) with drift, and reconstructs mine-level paths with volatility-scaled anchors. Its reported selection and hold-out diagnostics are inherited evidence, not revalidated forecasts or a jointly simulated cost state.



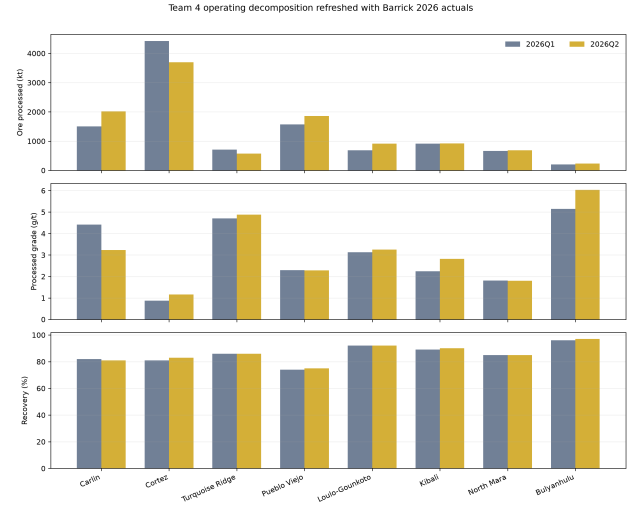
Source: Barrick Q1/Q2 2026 Mine Statistics. Selected Team 4 mine perimeter; company totals also include other operations.

Figure 1. Selected-mine Cost of Sales per ounce in Q1 and Q2 2026. Mine dispersion motivates preserving the Team 4 operating decomposition rather than replacing it with a single company-average cost.

The scenario uses 20 fixed quarterly cost entries, including two 2026 actuals and 18 forecast entries. It does not propagate the source study's operating distributions.

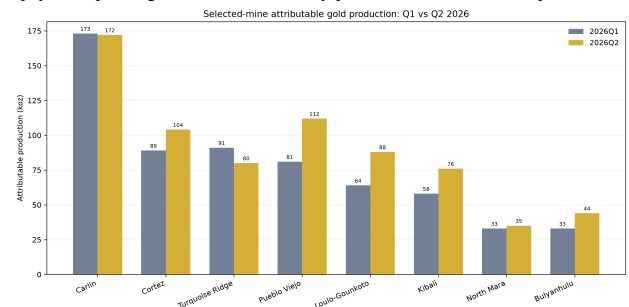
4.2 Production block

Production is reconstructed from ore processed, grade and recovery using the specifications in Section 3.4. This preserves mine heterogeneity. The SARIMA orders are admitted from the source study, which does not report their selection by AIC/BIC.



Ore processed, grade and recovery are physical operating drivers; no gold-price simulation enters this figure.

Figure 2. Ore processed, grade and recovery updated with Barrick 2026 actuals. These are physical operating drivers; no commodity-price simulation enters the panel.



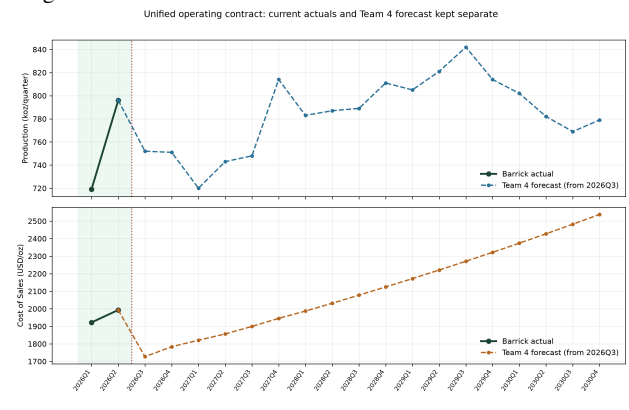
Source: Barrick Q1/Q2 2026 Mine Statistics. Selected Team 4 mine perimeter; company totals also include other operations.

Figure 3. Selected-mine attributable gold production in Q1 and Q2 2026. The current actuals reveal heterogeneous changes that are hidden by a company-level total.

The original study also develops Goodman variance propagation and confidence bands for the production components. Those distributions remain useful evidence on operating-model uncertainty, but they are not propagated jointly in the four-engine corporate experiment: only the refreshed point vectors are passed downstream.

4.3 Forecast handoff used by the valuation

The final handoff keeps observed and modelled quantities distinct. Barrick's Q1–Q2 2026 actual production and CoS establish the current operating state; the Team 4 forecasts begin at Q3 2026 and extend through the common valuation horizon.



The price layer is not shown: valuation gold paths are generated independently by the current Team 8 models.

Figure 4. Unified operating contract. Barrick Q1–Q2 2026 actuals are shown separately from the deterministic Team 4 production and CoS vectors used from Q3 2026 onward. The straight cost path is the conditional mean of a persistent-trend specification, not a claim that operating uncertainty disappears.

Prudential operating boundary

Option-implied laws supply conditional scenario shapes to the proxy layer. Costs retain a persistent trend without additional technology-driven cost reductions. This is prudential relative to otherwise identical efficiency-improvement scenarios, but it does not establish a mathematical lower bound. Stochastic costs could yield favourable and adverse outcomes; the current fixed vectors preserve attribution across gold engines. Growth is already included in the common DCF schedule.

5 CALIBRATION SAMPLE AUDIT

(a) September 2 implied-volatility surface

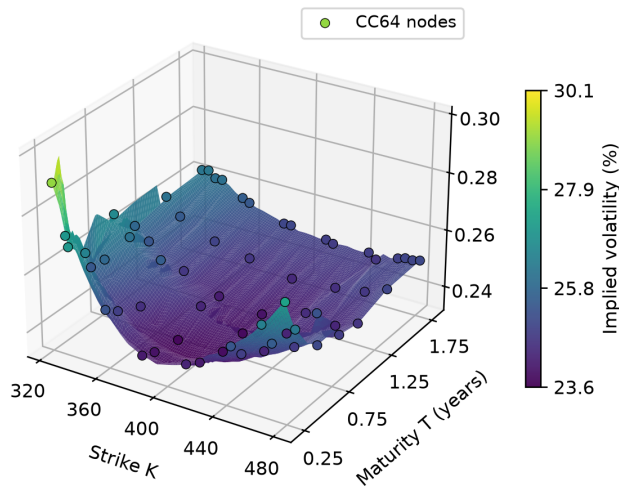


Figure 5. (a) GLD implied-volatility surface, 2 September 2026, with CC64 nodes. Interpolation is used for display only.

(b) Chebyshev–Chebyshev calibration nodes

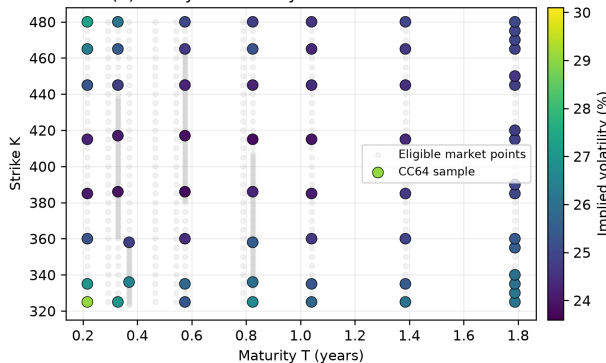


Figure 5 (continued). (b) The 64 actual contracts selected from 605 eligible calls across 12 expiries and 146 strikes.

5.1 Why structured sampling is retained

The option surface is irregular in strike and maturity. Team 8 therefore passes a fixed 8×8 Chebyshev–Chebyshev subset to the optimizer rather than allowing densely listed regions to dominate the loss. Every target is mapped to the nearest still-unused actual option: the 64-point calibration set contains no synthetic contracts. The figure audits the same 2 September parameter snapshot used by the refreshed valuation.

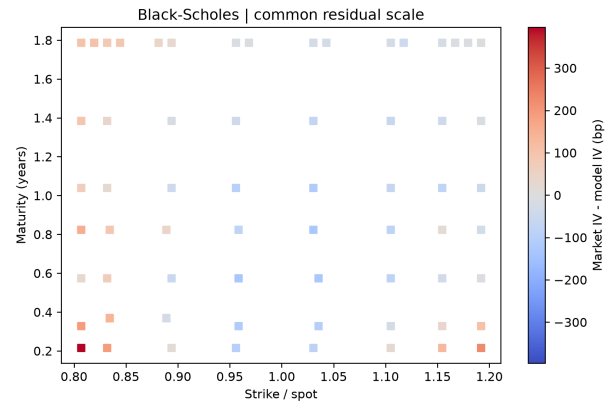
Evidence status

The calibrated inputs, residual maps and valuation paths share the same 2 September Team 8 parameter set. The U.S. Treasury curve is dated 2 September as well; the separate Q2 realized-gold level anchor remains explicit in the valuation contract.

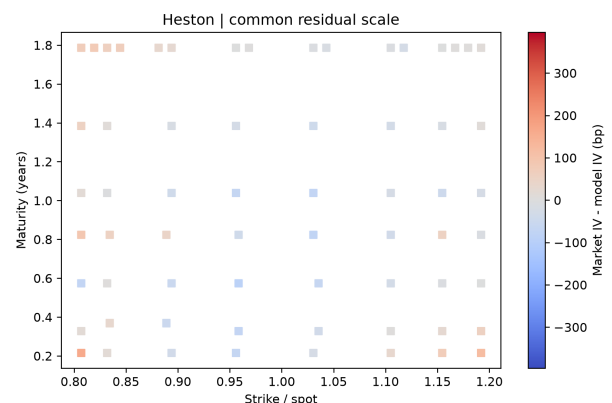
5.2 What the residual maps add

The residual heatmaps show where each specification systematically over- or under-prices the observed implied-volatility surface. Black–Scholes leaves the strongest smile and term-structure pattern. Table 2 compares current-surface fit; Audited Bates and Hawkes fits improve on the nested starts. Hawkes branching is 0.5594. All maps use one

colour scale; calibration fit does not establish economic superiority.

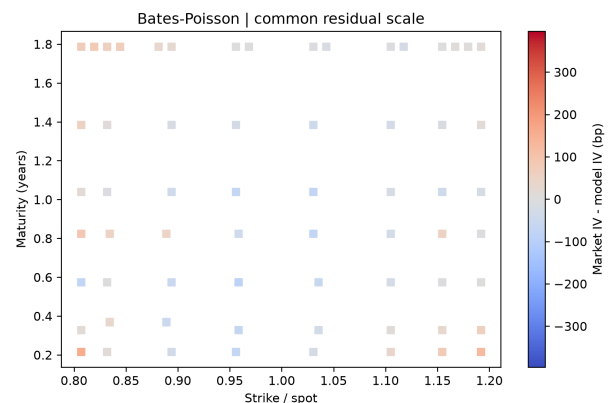


(a) Black–Scholes

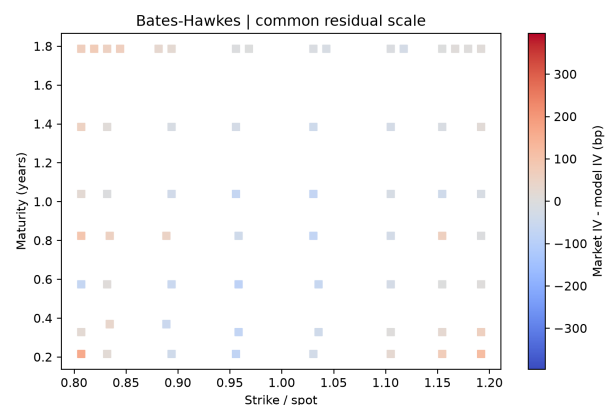


(b) Heston

Figure 6. Implied-volatility residual maps for the common 2 September CC64 sample. Blue denotes model IV above market IV and red denotes model IV below market; the audited fits retain feasible nested candidates.



(c) Bates–Poisson



(d) Full Bates–Hawkes

Figure 6 (continued). Bates–Poisson and Full Bates–Hawkes residual maps; all colourbars share the same residual range.

Table 2. Current-surface IV calibration RMSE.

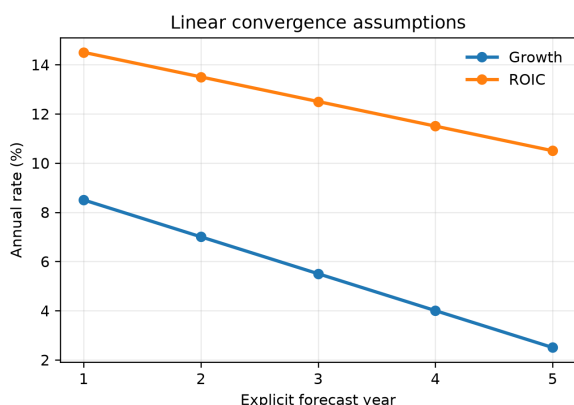
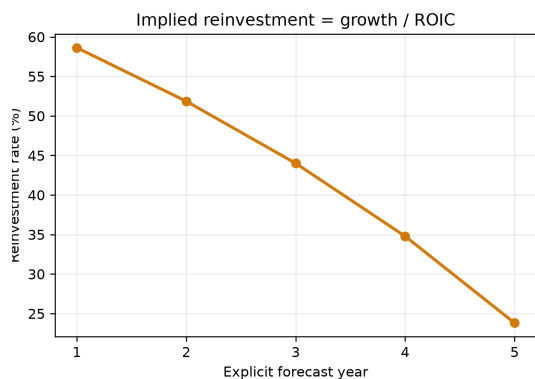
Model	RMSE (bp)
Black–Scholes	97.2107
Heston	49.4881
Bates–Poisson	49.4077
Full Bates–Hawkes	49.3258

6 DCF DRIVER AND TERMINAL-VALUE AUDIT

Growth, reinvestment, WACC and terminal closure are shared across gold engines. Table 3 summarizes the corporate assumptions.

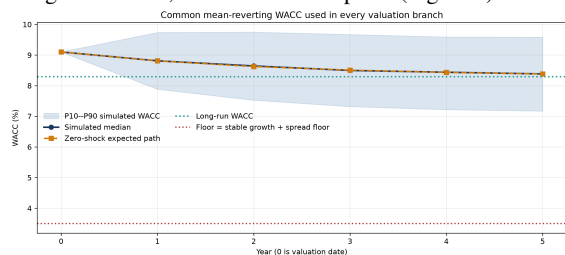
6.1 Growth, ROIC and reinvestment

Growth and ROIC converge linearly to their stable levels over five years. Reinvestment follows $b_t = g_t / ROIC_t$, linking growth to the capital required to sustain it (Figure 7).

**Figure 7.** Common explicit-period DCF schedule: growth and ROIC converge linearly over the five-year horizon.**Figure 7 (continued).** The implied reinvestment rate, defined as growth divided by ROIC, falls from 58.62% to 23.81%.

6.2 Common stochastic WACC

WACC follows the bounded mean-reverting process in Equation (17), with inputs in Table 3. Its annual innovation scale is $\sigma_w^* = \sigma_w \sqrt{(1 - e^{-2\kappa_w}) / (2\kappa_w)} = 0.7308$ percentage points before clipping. All engines reuse 8,192 annual shock paths (Figure 8).

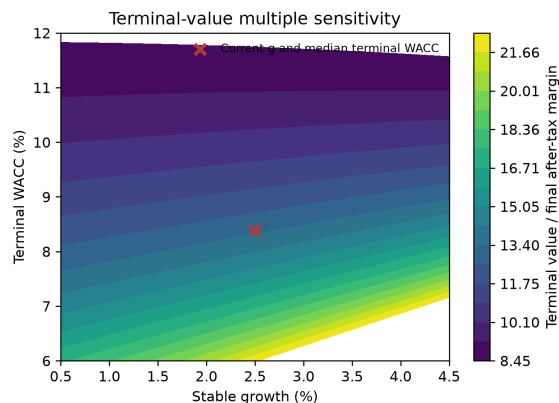


Editable fields: model.wacc_0, model.wacc_long_run, model.wacc_reversion, model.wacc_volatility and model.terminal_spread_floor. All four gold models reuse the same 8,192-path WACC shock array.

Figure 8. Common mean-reverting WACC. The band is the P10–P90 range from the shared 8,192-path shock array; the orange line is the zero-shock path and the dotted lines show the long-run level and admissibility floor.

6.3 Terminal closure and sensitivity

The terminal block uses $g_\infty = 2.50\%$, $ROIC_\infty = 10.50\%$ and terminal reinvestment of 23.81%. At the current median terminal WACC of 8.384%, the normalized terminal multiple is 13.27× final after-tax margin. The spread floor enforces $WACC_T > g_\infty$. No simulated rate reaches either clipping bound in the base run.

**Figure 9.** Terminal-value multiple sensitivity to stable growth and terminal WACC. The cross marks the configured stable-growth rate and current median terminal WACC.**Figure 9 (continued).** Median proxy under three terminal conventions. Five-year-only values are about USD 13 billion versus about USD 60 billion with perpetuity. Zero terminal is a truncation scenario, not an estimated reserve life.**Table 3.** Editable common DCF inputs.

Parameter	Value
Tax rate	25.00%
$g_1 \rightarrow g_\infty$	8.50–2.50%
$ROIC_1 \rightarrow ROIC_\infty$	14.50–10.50%
$WACC_0 \rightarrow \bar{w}$	9.10–8.30%
WACC reversion	0.45
WACC volatility	0.90%
WACC floor/cap	3.50/25.00%
Terminal reinvestment	23.81%

Interpretation

These are methodological assumptions, not filing-calibrated Barrick estimates. Terminal closure dominates the reported operating proxy. The legacy positive terminal increases the mean by USD 1.14–1.49 billion relative to signed closure.

The quarterly bridge multiplies simulated gold price less unit Cost of Sales, both in USD per ounce, by production in thousand ounces, then divides by 1,000 to obtain component margin in USD millions. Four quarterly margins are summed before annual taxes, reinvestment and discounting. This proxy is not reported EBIT or FCF: production must be reconciled with sales and inventory, and depreciation, capital expenditure and working capital must enter without double counting. Fixed operating vectors exclude mine-level shocks and price-cost dependence.

7 OPERATING-PROXY SENSITIVITIES

Table 4. Audited Bates–Hawkes operating proxy.

Metric	USD billion or %
P10	1.03
Median (P50)	59.82
P90	172.60
Mean	77.95
Negative proxy paths	9.30%
Five-year-only median	12.89
Legacy floor mean uplift	1.49

Interpretation

These are unconstrained aggregate operating proxies, not enterprise or equity values. The Q-law/WACC combination has no demonstrated corporate pricing interpretation. Negative outcomes are not negative limited-liability share prices; no market comparison is made.

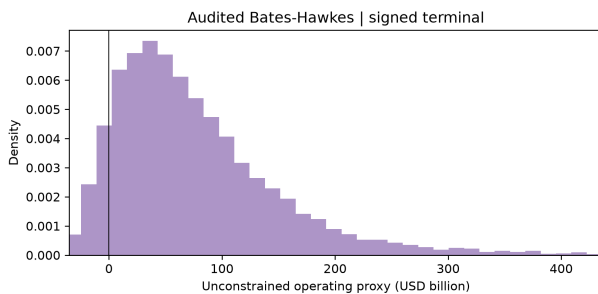


Figure 10. Signed-terminal operating proxy under audited Bates–Hawkes parameters. The common contract is reused across 8,192 paths. Display limits are the 0.5th–99.5th percentiles; the full tails remain in the data. This is not a confidence interval or physical probability law.

7.1 Cross-model central tendency

Audited parameter refinements dated 7 September use the unchanged 2 September market sample. Cross-model medians range from USD 59.12 to 60.10 billion and P90 from USD 172.60 to 174.02 billion. Their proximity should not be interpreted as a robust ranking of companies or models.

Table 5. Aggregate proxy medians, USD billion.

Gold engine	Median	Five-year only
Black–Scholes	60.10	13.17
Heston	59.77	13.08
Bates–Poisson	59.12	12.83
Full Bates–Hawkes	59.82	12.89

The simulation spread describes conditional scenario dispersion; Monte Carlo error concerns numerical estimates. Doubling paths to 16,384, doubling fine steps to 520 and changing the seed gives Bates–Hawkes medians of 59.34, 61.59 and 61.38 billion versus 59.82 at baseline. Such numerical variation exceeds some cross-model median gaps. Negative values expose the unconstrained mapping, not negative limited-liability shares. The data retain these outcomes rather than silently flooring them.

7.2 Structural illustration, not model selection

Bates–Hawkes illustrates the richest law, not a preferred corporate valuation. Its audited branching ratio is 0.5594 and kernel half-life $\log(2)/\beta = 0.2522$ years. These are risk-neutral fitted parameters, not measured geopolitical persistence. The earlier OOS exercise below uses the original, unrefined origin fits.

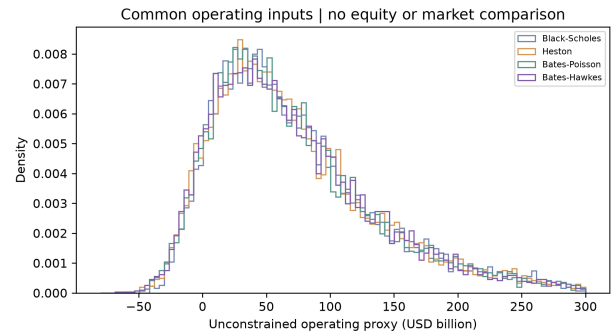


Figure 11. Audited operating-proxy distributions under four gold-price laws. Operations, WACC and signed terminal closure are common. No share count, equity bridge or market reference is used.

8 OUT-OF-SAMPLE EVIDENCE AND MODEL RISK

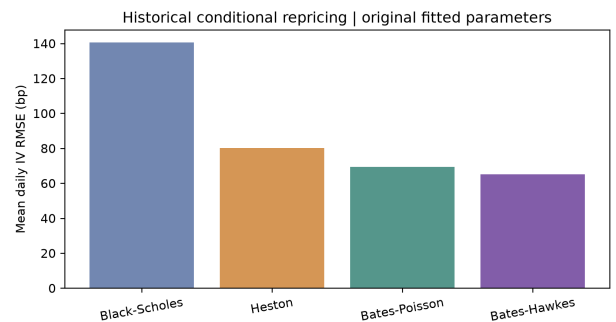


Figure 12. Dense-only rolling evidence on 30 forecast origins and 4,667 common target observations. The original fits give Bates–Hawkes the lowest mean daily IV RMSE.

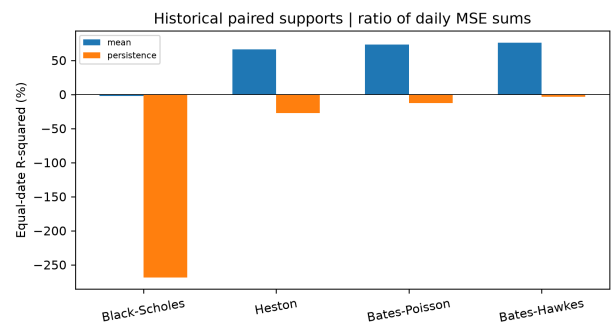


Figure 12 (continued). Positive R_{OOS}^2 indicates improvement over the stated benchmark.

8.1 Conditional next-date repricing

Model comparisons use 4,667 targets; persistence comparisons use the 4,026 inside the origin interpolation domain. Benchmark scores therefore have different supports; equal date weighting limits the influence of densely populated dates.

Reported RMSE is the mean of daily RMSEs, not the root of mean daily MSE used by R^2 . Values are 140.6528 bp for Black–Scholes, 80.1964 bp for Heston, 69.3055 bp for Bates and 65.0507 bp for Full Bates–Hawkes. Relative to the origin mean-IV benchmark, the corresponding R_{OOS}^2 values are -2.58% , 66.07% , 72.97% and 75.81% . None beats interpolated IV-surface persistence, but Full Bates–Hawkes is closest at -3.68% , followed by Bates at -12.92% and Heston at -27.24% .

Model-risk conclusion

Historical OOS uses the original fits; audited current fits are a separate experiment. A paired five-date block bootstrap gives a Hawkes-minus-Bates MSE interval $[-8.95, -2.77] \times 10^{-6}$, exploratory with only 30 dates.

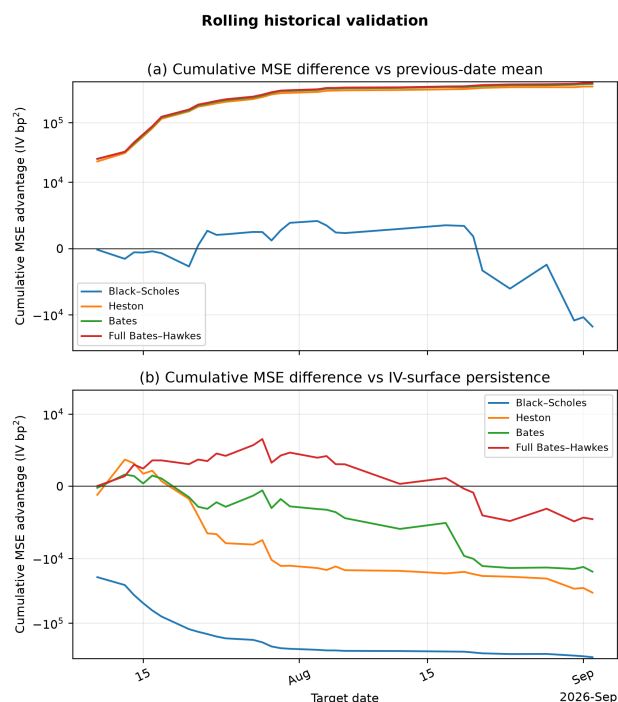


Figure 13. Cumulative rolling MSE differences against the origin mean IV and interpolated IV-surface persistence. Upward movement favours the option model; the stronger persistence benchmark remains difficult to beat.

8.2 Jump-layer interpretation

The self-exciting extension is motivated by clustered option-implied jump activity. Relative to the constant Poisson baseline, the Hawkes specification permits intensity bursts followed by endogenous decay. The branching ratio and decay parameter have different meanings; fitted risk-neutral self-excitation is not evidence of physical news clustering.

Without a dated news filtration or parameter-stability evidence, these option-implied bursts cannot be identified as causal responses to geopolitical events.

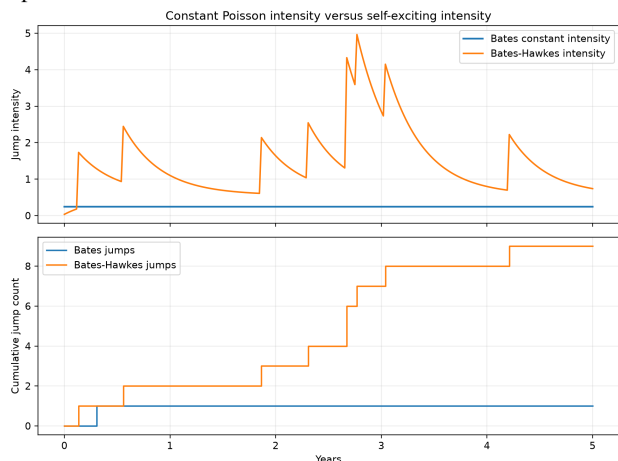


Figure 14. Illustrative simulated Poisson and Hawkes paths under the original parameter set. A single realization cannot establish clustering in observed data.

8.3 Interpretive limitations and validation

Reproducibility, numerical accuracy and economic validity are distinct. A good option fit cannot validate the GLD-to-gold mapping or the Q-law/WACC combination. The experiment identifies sensitivity to a chosen numerical contract, not corporate fair value.

The old run equated enterprise and equity proxies using zero bridge placeholders. This revision removes that share-price interpretation; debt, cash, minority claims and copper remain unreconciled. A genuine

valuation must start from one dated filing and the matching listed security, establish ownership and attributable production, and distinguish component margin from reported operating profit. Depreciation, cash taxes, working capital, sustaining capital expenditure and growth investment must then enter consistently, with no deduction counted twice. Net debt, non-operating assets, streams and other claims must be matched to a diluted share count on the same basis. Only then can a per-share output acquire a filing-reconciled interpretation.

8.4 Operating uncertainty and research priorities

Production, grade, recovery, mine mix and costs may depend on gold prices, inflation and exchange rates. Independence is not automatic: operating dependence and Goodman variance propagation require explicit assumptions. Technology scenarios must include implementation costs and investment; a smooth conditional cost mean does not imply absence of uncertainty.

A mining-company terminal block needs an explicit resource interpretation. Continued production may require reserve conversion, replacement investment or new projects, while depletion and closure liabilities can reduce the duration and value of the cash-flow stream. A positive WACC-growth spread controls a mathematical denominator but does not establish that a perpetual operating horizon is economically justified. Finite-resource and going-concern closures should be compared under coherent investment assumptions before attributing differences entirely to the gold-price model. Such a comparison is a future validation exercise, not a result of the present run.

Market maturities end at 653 days, well before five scenario years and perpetuity. Fixed parameters beyond that horizon are an extrapolation assumption, not market-identified information. The five-year-only comparison removes terminal continuation but does not calibrate depletion, replacement investment or closure costs. A bucketed term-structure and finite-reserve model remain necessary for an economic valuation.

CONCLUSION

Holding operations and corporate assumptions fixed isolates the effect of gold-price model choice. Nested-candidate safeguards improve the current Bates and Hawkes fits. Historical OOS remains distinct and no original engine beats interpolated persistence. Numerical and terminal sensitivity prevent a corporate value ranking.

The persistent-cost assumption is prudential only relative to otherwise identical efficiency-improvement scenarios, not a lower bound. The contribution is an auditable interdisciplinary framework, not a certified value: commodity basis, probability measure, operating uncertainty, corporate accounting and finite-resource closure remain the priorities for economic validation.

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